

🎯 Maths Unit 3, Part 3D: Ordered pairs, Cartesian product aur Relations

Aasan Notes for Seemab · kitab ke page 60 se 66 tak · practice: Exercise 3.3 aur Exercise 3.4

Pehle ek sawal. Do jama paanch barabar saat, aur paanch jama do bhi barabar saat. Dono ek jaise. Lekin agar main likhoon **(2, 5)** aur **(5, 2)**, to ye do bilkul alag cheezein hain. Aisa kyun? Zara socho, jawab is note ke akhir mein khud de dogi.

📌 1. Ordered Pair (p60)

📖 Kitab ki definition

“Pairs of two numbers in which order of numbers is not invertible, is called an ordered pair.” Numbers **chhote bracket** mein likhe jate hain aur beech mein **comma** hota hai.

Jaise **(a, 5)** mein **a** ko *first element* kehte hain aur **5** ko *second element*. Elements ki jagah badal do to poora ordered pair hi badal jata hai.

🧠 Isay yaad rakhne ka asaan tareeqa

Ghar ka pata socho: **gali number 2, makaan number 5**. Ab ulta kar do: gali 5, makaan 2. Bilkul doosra ghar, doosre mohalle mein. Number wohi do hain, magar **tarteeb** badalte hi jagah badal gayi.

Kitab yehi baat geometry se kehti hai: “As in geometry, position of a point is determined by ordered pair, therefore (2, 5) and (5, 2) represent two different points.” Yani **(2, 5) ≠ (5, 2)**.

📖 Equality of Two Ordered Pairs

Do ordered pairs **(a, b)** aur **(c, d)** tabhi barabar hote hain jab: **a = c aur b = d**. Yani pehla pehle ke barabar, doosra doosre ke barabar.

📌 Example 15 (p60)

Find the values of **x** and **y** when **(x - 3y, 5x + 1) = (4, 6)**.

Solution. Equality of ordered pairs se do alag equations bante hain:

$$x - 3y = 4 \text{ (i) aur } 5x + 1 = 6 \text{ (ii)}$$

$$\text{(ii) se: } 5x = 6 - 1 = 5, \text{ to } x = 1.$$

$$\text{Ab } x = 1 \text{ ko (i) mein rakho: } 1 - 3y = 4 \rightarrow -3y = 3 \rightarrow y = -1.$$

Do unknowns ke liye do equations: yehi is sawal ki poori chaal hai.

✂ 2. Cartesian Product of Sets (p61, p62)

📖 Kitab ki definition

Agar A aur B do non-empty sets hain to Cartesian product: **A × B = {(x, y) | x ∈ A ∧ y ∈ B}**

Aur isi tarah **B × A = {(y, x) | y ∈ B ∧ x ∈ A}**.

Matlab: A ka **har** element B ke **har** element ke saath ek jora banata hai. Canteen ka menu socho: teen tarah ke sandwich aur do drinks. Har sandwich har drink ke saath chal sakta hai, to kul **3 × 2 = 6** alag deals banti hain. Bas yehi Cartesian product hai.

📌 Kitab ki misal (p61)

$$A = \{0, 1, 2\}, B = \{3, 4\}$$

$$A \times B = \{(0,3), (0,4), (1,3), (1,4), (2,3), (2,4)\} \text{ (6 pairs)}$$

$$B \times A = \{(3,0), (3,1), (3,2), (4,0), (4,1), (4,2)\} \text{ (6 pairs)}$$

Ghor karo: dono mein 6 pairs hain, magar **A × B aur B × A ek jaise nahin hain**, kyunke har jore ki tarteeb ulta hai.

Teen pakke formulay (Key Facts, p62)

- $n(A \times B) = n(A) \times n(B)$ Pairs ginne ke liye elements multiply kar do, likhne ki zaroorat nahin.
- $A \times B = \emptyset$ agar A ya B mein se koi bhi khaali ho.
- $A \times B$ ke subsets ki tadaad = $2^{n(A \times B)}$

3. Binary Relation (p63)

Kitab ki definition

"A binary relation R in the set $A \times B$ is a subset of the Cartesian product $A \times B$."

Yani R relation hai **agar aur sirf agar** $R \subseteq A \times B$.

Cartesian product **saaray** mumkin joray banata hai. Relation un mein se sirf **woh joray chunta hai jo kisi shart par poore utarte hain**. Samajhne ka tareeqa: $A \times B$ poori class ki list hai, aur R woh list hai jis par likha ho "sirf woh larkiyan jo cricket khelti hain". R hamesha $A \times B$ ke andar hi rahegi.

Relations ginne ka formula

Number of binary relations in $A \times B = 2^{n(A \times B)}$

Wajah: har jora ya to R mein hoga ya nahin hoga, yani har jore par do choices.

Example 17 aur 18 (p63)

17. $A = \{5, 10, 15, 20, 25\}$. $n(A) = 5$, to $n(A \times A) = 5 \times 5 = 25$.

Number of binary relations in $A = 2^{25}$.

18. $P = \{2, 3\}$, Q mein bhi 2 elements. $n(P \times Q) = 2 \times 2 = 4$.

Number of binary relations = $2^4 = 16$.

Example 19 (p63), shart lagana

$A = \{1, 2, 3\}$, $B = \{0, 1, 3\}$, aur $R = \{(a, b) \mid a \in A, b \in B \text{ aur } a > b\}$.

Pehle poora $A \times B$ likho (9 joray), phir un mein se sirf woh chuno jahan pehla number doosre se bara ho:

$R = \{(1,0), (2,0), (2,1), (3,0), (3,1)\}$

Tareeqa hamesha yehi: pehle saaray joray, phir shart ki chhanni.

4. Domain aur Range (p64, p65)

Yaad rakhne ka usool

Domain of R = saaray **pehle** elements ka set. **Range of R** = saaray **doosray** elements ka set.

Do lafz jo kabhi nahin bhoologi

Domain = Daayen wala nahin, **pehla** wala. Sochne ka asaan tareeqa: ordered pair (**input**, **output**) hai. Jo andar jata hai wo **Domain**, jo bahar aata hai wo **Range**.

Example 20(c) (p65)

$E = \{2, 4, 6\}$, $F = \{0, 1, 2\}$, $R_2 = \{(x, y) \mid x \in E, y \in F \text{ aur } x + y = 6\}$

$E \times F$ ke 9 joray banao, phir sirf woh rakho jinka total 6 ho:

$R_2 = \{(4, 2), (6, 0)\}$

Dom $R_2 = \{4, 6\}$ aur Range $R_2 = \{0, 2\}$

5. Inverse Relation (p65, p66)

Kitab ki definition

Agar $R = \{(a, b) \mid a \in A, b \in B\}$ ek relation hai A se B tak, to iska inverse:

$$R^{-1} = \{(b, a) \mid b \in B, a \in A\}$$

Yani **har jore ko ulta likh do**. Bas itni si baat.

Do natije jo imtihan mein aksar poochhe jate hain

$$\text{Dom } R^{-1} = \text{Range } R \quad \text{aur} \quad \text{Range } R^{-1} = \text{Dom } R$$

Samajhna asaan hai: jab har jora ulta ho gaya to jo pehle bahar aa raha tha wo ab andar ja raha hai.

Sab se aam ghalti

(a, b) ko (b, a) likh dena jab sawal ne kaha hi nahin. Inverse sirf tab lo jab sawal mein R^{-1} ya lafz **inverse** likha ho. Aur Key Fact yaad rakho (p62): "The ordered pair (a, b) cannot be written as (b, a) ."

Ab shuru wala sawal

Do jama paanch aur paanch jama do barabar hain kyunke **jorne mein tarteeb ki koi ahmiyat nahin**. Lekin $(2, 5)$ aur $(5, 2)$ jama nahin hain, ye **jagah** bata rahe hain: ek gali 2 makaan 5, doosra gali 5 makaan 2. Isi liye kitab kehti hai ke ye do alag points hain. **Jahan tarteeb maayne rakhti ho, wahan bracket aur comma lagta hai.**

Quiz

Ye teeno sawal tumhare Maths paper mein bhi aayenge, is liye inhein achhi tarah karo.

- Find the values of x and y when $(2x + 1, y - 4) = (7, 3)$. Pehle equality of ordered pairs ka usool likho, phir dono equations hal karo.
- Let $A = \{1, 2\}$ and $B = \{3, 4, 5\}$. (i) Write $A \times B$ in tabular form. (ii) Using the formula, state $n(A \times B)$. (iii) How many binary relations are there in $A \times B$?
- Let $C = \{2, 4, 6\}$, $D = \{4, 6, 8\}$ and $R = \{(x, y) \mid x \in C, y \in D \wedge y = x + 2\}$. Write R in tabular form, then find **Dom R**, **Range R** and R^{-1} .